

This file presents p-values for comparing pairs of statistical estimates, where each pair consists of two distinct measures: the ratio of predicted contrasts from a GEE model examining the effect of maternal SES on HAZ & WAZ, and the ratio of beta coefficients from a regression model predicting BMI-Z as a function of WAZ and HAZ. Each estimate is accompanied by its 95% confidence interval. The p-values are calculated to test three hypotheses: whether the first estimate is significantly lower than, greater than, or equal to the second estimate.

Steps for Comparison

1. Calculate the Standard Errors (SE)

For each estimate, SE is calculated from the 95% CI as:

$$SE = \frac{Upper\ Bound - Lower\ Bound}{2 \times 1.96}$$

2. Calculate the Difference and SE_diff

- Difference between the estimates:

$$Difference = \beta_1 - \beta_2$$

- SE_diff (Standard Error of the Difference):

$$SE_{diff} = \sqrt{SE_1^2 + SE_2^2}$$

3. Calculate the Z-Score

$$Z = \frac{Difference}{SE_{diff}}$$

4. Find the p-Values

- $H_1 <$ (Estimate 1 < Estimate 2):

$$p = CDF(Z)$$

- $H_1 >$ (Estimate 1 > Estimate 2):

$$p = 1 - CDF(Z)$$

- $H_1 =$ (Estimate 1 $\neq$ Estimate 2):

$$p = 2 \times CDF(-|Z|)$$

Pair Calculations

Pair 1

- Estimate 1: 0.502, CI: (-1.120, 2.123)
- Estimate 2: 1.257, CI: (1.207, 1.307)

Standard Errors:

$$SE_1 = \frac{2.123 - (-1.120)}{2 \times 1.96} \approx \frac{3.243}{3.92} \approx 0.828$$

$$SE_2 = \frac{1.307 - 1.207}{2 \times 1.96} \approx \frac{0.100}{3.92} \approx 0.025$$

Difference and SE_diff:

$$Difference = 0.502 - 1.257 = -0.755$$

$$SE_{diff} = \sqrt{0.828^2 + 0.025^2} \approx \sqrt{0.685 + 0.0006} \approx 0.828$$

Z-Score:

$$Z = \frac{-0.755}{0.828} \approx -0.912$$

p-Values:

- $H_1 <$ (Estimate 1 < Estimate 2): $p \approx 0.179$
- $H_1 >$ (Estimate 1 > Estimate 2): $p \approx 0.821$
- $H_1 =$ (Estimate 1 $\neq$ Estimate 2): $p \approx 0.358$

2. Pair 2

- Estimate 1: 0.571, CI: (-1.425, 2.567)
- Estimate 2: 1.295, CI: (1.242, 1.347)

Standard Errors:

$$SE_1 = \frac{2.567 - (-1.425)}{2 \times 1.96} \approx 0.994$$

$$SE_2 = \frac{1.347 - 1.242}{2 \times 1.96} \approx 0.027$$

Difference and SE_diff:

$$Difference = 0.571 - 1.295 = -0.724$$

$$SE_{diff} = \sqrt{0.994^2 + 0.027^2} \approx 0.994$$

Z-Score:

$$Z = \frac{-0.724}{0.994} \approx -0.728$$

p-Values:

- $H_1 <$ (Estimate 1 < Estimate 2): $p \approx 0.233$
- $H_1 >$ (Estimate 1 > Estimate 2): $p \approx 0.767$
- $H_1 =$ (Estimate 1 $\neq$ Estimate 2): $p \approx 0.466$

3. Pair 3

- Estimate 1: 0.617, CI: (-1.583, 2.816)
- Estimate 1: 0.617, CI: (-1.583, 2.816)
- Estimate 2: 1.291, CI: (1.237, 1.344)

Standard Errors:

$$SE_1 = \frac{2.816 - (-1.583)}{2 \times 1.96} \approx 1.186$$

$$SE_2 = \frac{1.344 - 1.237}{2 \times 1.96} \approx 0.027$$

Difference and SE_diff:

$$Difference = 0.617 - 1.291 = -0.674$$

$$SE_{diff} = \sqrt{1.186^2 + 0.027^2} \approx 1.186$$

Z-Score:

$$Z = \frac{-0.674}{1.186} \approx -0.568$$

p-Values:

- $H_1 <$ (Estimate 1 < Estimate 2): $p \approx 0.285$
- $H_1 >$ (Estimate 1 > Estimate 2): $p \approx 0.715$
- $H_1 =$ (Estimate 1 $\neq$ Estimate 2): $p \approx 0.570$

4. Pair 4

- Estimate 1: 0.697, CI: (-1.832, 3.224)
- Estimate 2: 1.273, CI: (1.214, 1.331)

Standard Errors:

$$SE_1 = \frac{3.224 - (-1.832)}{2 \times 1.96} \approx 1.720$$

$$SE_2 = \frac{1.331 - 1.214}{2 \times 1.96} \approx 0.027$$

Difference and SE_diff:

$$Difference = 0.697 - 1.273 = -0.576$$

$$SE_{diff} = \sqrt{1.720^2 + 0.027^2} \approx 1.720$$

Z-Score:

$$Z = \frac{-0.576}{1.720} \approx -0.335$$

p-Values:

- $H_1 <$ (Estimate 1 < Estimate 2): $p \approx 0.370$
- $H_1 >$ (Estimate 1 > Estimate 2): $p \approx 0.630$
- $H_1 =$ (Estimate 1 $\neq$ Estimate 2): $p \approx 0.740$

5. Pair 5

- Estimate 1: 0.762, CI: (-1.838, 3.361)
- Estimate 2: 1.286, CI: (1.201, 1.372)

Standard Errors:

$$SE_1 = \frac{3.361 - (-1.838)}{2 \times 1.96} \approx 1.734$$

$$SE_2 = \frac{1.372 - 1.201}{2 \times 1.96} \approx 0.087$$

Difference and SE_diff:

$$Difference = 0.762 - 1.286 = -0.524$$

$$SE_{diff} = \sqrt{1.734^2 + 0.087^2} \approx 1.734$$

Z-Score:

$$Z = \frac{-0.524}{1.734} \approx -0.302$$

p-Values:

- $H_1 <$ (Estimate 1 < Estimate 2): $p \approx 0.382$
- $H_1 >$ (Estimate 1 > Estimate 2): $p \approx 0.618$
- $H_1 =$ (Estimate 1 $\neq$ Estimate 2): $p \approx 0.764$

6. Pair 6

- Estimate 1: 0.592, CI: (-0.924, 2.108)
- Estimate 2: 1.256, CI: (1.209, 1.303)

Standard Errors:

$$SE_1 = \frac{2.108 - (-0.924)}{2 \times 1.96} \approx 1.558$$

$$SE_2 = \frac{1.303 - 1.209}{2 \times 1.96} \approx 0.025$$

Difference and SE_diff:

$$Difference = 0.592 - 1.256 = -0.664$$

$$SE_{diff} = \sqrt{1.558^2 + 0.025^2} \approx 1.558$$

Z-Score:

$$Z = \frac{-0.664}{1.558} \approx -0.426$$

p-Values:

- $H_1 <$ (Estimate 1 < Estimate 2): $p \approx 0.335$
- $H_1 >$ (Estimate 1 > Estimate 2): $p \approx 0.665$
- $H_1 =$ (Estimate 1 $\neq$ Estimate 2): $p \approx 0.670$

7. Pair 7

- Estimate 1: 0.589, CI: (-1.202, 2.379)
- Estimate 2: 1.271, CI: (1.224, 1.318)

Standard Errors:

$$SE_1 = \frac{2.379 - (-1.202)}{2 \times 1.96} \approx 1.732$$

$$SE_2 = \frac{1.318 - 1.224}{2 \times 1.96} \approx 0.025$$

Difference and SE_diff:

$$Difference = 0.589 - 1.271 = -0.682$$

$$SE_{diff} = \sqrt{1.732^2 + 0.025^2} \approx 1.732$$

Z-Score:

$$Z = \frac{-0.682}{1.732} \approx -0.394$$

p-Values:

- $H_1 <$ (Estimate 1 < Estimate 2): $p \approx 0.347$
- $H_1 >$ (Estimate 1 > Estimate 2): $p \approx 0.653$
- $H_1 =$ (Estimate 1 $\neq$ Estimate 2): $p \approx 0.694$

8. Pair 8

- Estimate 1: 0.588, CI: (-1.221, 2.397)
- Estimate 2: 1.261, CI: (1.212, 1.310)

Standard Errors:

$$SE_1 = \frac{2.397 - (-1.221)}{2 \times 1.96} \approx 1.741$$

$$SE_2 = \frac{1.310 - 1.212}{2 \times 1.96} \approx 0.025$$

Difference and SE_diff:

$$Difference = 0.588 - 1.261 = -0.673$$

$$SE_{diff} = \sqrt{1.741^2 + 0.025^2} \approx 1.741$$

Z-Score:

$$Z = \frac{-0.673}{1.741} \approx -0.387$$

p-Values:

- $H_1 <$ (Estimate 1 < Estimate 2): $p \approx 0.350$
- $H_1 >$ (Estimate 1 > Estimate 2): $p \approx 0.650$
- $H_1 =$ (Estimate 1 $\neq$ Estimate 2): $p \approx 0.700$

9. Pair 9

- Estimate 1: 0.582, CI: (-1.293, 2.457)
- Estimate 2: 1.263, CI: (1.209, 1.316)

Standard Errors:

$$SE_1 = \frac{2.457 - (-1.293)}{2 \times 1.96} \approx 1.380$$

$$SE_2 = \frac{1.316 - 1.209}{2 \times 1.96} \approx 0.027$$

Difference and SE_diff:

$$Difference = 0.582 - 1.263 = -0.681$$

$$SE_{diff} = \sqrt{1.380^2 + 0.027^2} \approx 1.380$$

Z-Score:

$$Z = \frac{-0.681}{1.380} \approx -0.494$$

p-Values:

- $H_1 < (\text{Estimate 1} < \text{Estimate 2}): p \approx 0.312$
- $H_1 > (\text{Estimate 1} > \text{Estimate 2}): p \approx 0.688$
- $H_1 = (\text{Estimate 1} \neq \text{Estimate 2}): p \approx 0.624$

10. Pair 10

- Estimate 1: 0.577, CI: (-1.358, 2.511)
- Estimate 2: 1.257, CI: (1.164, 1.350)

Standard Errors:

$$SE_1 = \frac{2.511 - (-1.358)}{2 \times 1.96} \approx 1.916$$

$$SE_2 = \frac{1.350 - 1.164}{2 \times 1.96} \approx 0.095$$

Difference and SE_diff:

$$Difference = 0.577 - 1.257 = -0.680$$

$$SE_{diff} = \sqrt{1.916^2 + 0.095^2} \approx 1.916$$

Z-Score:

$$Z = \frac{-0.680}{1.916} \approx -0.354$$

p-Values:

- $H_1 < (\text{Estimate 1} < \text{Estimate 2}): p \approx 0.362$
- $H_1 > (\text{Estimate 1} > \text{Estimate 2}): p \approx 0.638$
- $H_1 = (\text{Estimate 1} \neq \text{Estimate 2}): p \approx 0.724$

11. Pair 11

- Estimate 1: 2.434, CI: (-2.496, 7.364)
- Estimate 2: 1.397, CI: (1.349, 1.445)

Standard Errors:

$$SE_1 = \frac{7.364 - (-2.496)}{2 \times 1.96} \approx 4.986$$

$$SE_2 = \frac{1.445 - 1.349}{2 \times 1.96} \approx 0.025$$

Difference and SE_diff:

$$Difference = 2.434 - 1.397 = 1.037$$

$$SE_{diff} = \sqrt{4.986^2 + 0.025^2} \approx 4.986$$

Z-Score:

$$Z = \frac{1.037}{4.986} \approx 0.208$$

p-Values:

- $H_1 <$ (Estimate 1 < Estimate 2): $p \approx 0.418$
- $H_1 >$ (Estimate 1 > Estimate 2): $p \approx 0.582$
- $H_1 =$ (Estimate 1 $\neq$ Estimate 2): $p \approx 0.836$

12. Pair 12

- Estimate 1: 1.742, CI: (-2.639, 6.117)
- Estimate 2: 1.399, CI: (1.352, 1.447)

Standard Errors:

$$SE_1 = \frac{6.117 - (-2.639)}{2 \times 1.96} \approx 4.487$$

$$SE_2 = \frac{1.447 - 1.352}{2 \times 1.96} \approx 0.025$$

Difference and SE_diff:

$$Difference = 1.742 - 1.399 = 0.343$$

$$SE_{diff} = \sqrt{4.487^2 + 0.025^2} \approx 4.487$$

Z-Score:

$$Z = \frac{0.343}{4.487} \approx 0.076$$

p-Values:

- $H_1 < (\text{Estimate 1} < \text{Estimate 2}): p \approx 0.531$
- $H_1 > (\text{Estimate 1} > \text{Estimate 2}): p \approx 0.469$
- $H_1 = (\text{Estimate 1} \neq \text{Estimate 2}): p \approx 0.938$

Pair 13

- Estimate 1: 1.362, CI: (-2.277, 5.998)
- Estimate 2: 1.390, CI: (1.340, 1.439)

Standard Errors:

$$SE_1 = \frac{5.998 - (-2.277)}{2 \times 1.96} \approx \frac{8.275}{3.92} \approx 2.108$$

$$SE_2 = \frac{1.439 - 1.340}{2 \times 1.96} \approx \frac{0.099}{3.92} \approx 0.025$$

Difference and SE_diff:

$$\text{Difference} = 1.362 - 1.390 = -0.028$$

$$SE_{\text{diff}} = \sqrt{2.108^2 + 0.025^2} \approx \sqrt{4.446 + 0.001} \approx 2.110$$

Z-Score:

$$Z = \frac{-0.028}{2.110} \approx -0.013$$

p-Values:

- $H_1 < (\text{Estimate 1} < \text{Estimate 2}): p \approx 0.495$
- $H_1 > (\text{Estimate 1} > \text{Estimate 2}): p \approx 0.505$

- $H_1 = (\text{Estimate 1} \neq \text{Estimate 2}): p \approx 0.990$

Pair 14

- Estimate 1: 0.834, CI: (-1.245, 2.914)
- Estimate 2: 1.372, CI: (1.317, 1.426)

Standard Errors:

$$SE_1 = \frac{2.914 - (-1.245)}{2 \times 1.96} \approx \frac{4.159}{3.92} \approx 1.061$$

$$SE_2 = \frac{1.426 - 1.317}{2 \times 1.96} \approx \frac{0.109}{3.92} \approx 0.028$$

Difference and SE_diff:

$$\text{Difference} = 0.834 - 1.372 = -0.538$$

$$SE_{\text{diff}} = \sqrt{1.061^2 + 0.028^2} \approx \sqrt{1.127 + 0.001} \approx 1.061$$

Z-Score:

$$Z = \frac{-0.538}{1.061} \approx -0.508$$

p-Values:

- $H_1 < (\text{Estimate 1} < \text{Estimate 2}): p \approx 0.306$
- $H_1 > (\text{Estimate 1} > \text{Estimate 2}): p \approx 0.694$
- $H_1 = (\text{Estimate 1} \neq \text{Estimate 2}): p \approx 0.612$

Pair 15

- Estimate 1: 0.487, CI: (-1.043, 2.017)
- Estimate 2: 1.397, CI: (1.349, 1.445)

Standard Errors:

$$SE_1 = \frac{2.017 - (-1.043)}{2 \times 1.96} \approx \frac{3.060}{3.92} \approx 0.781$$

$$SE_2 = \frac{1.445 - 1.349}{2 \times 1.96} \approx \frac{0.096}{3.92} \approx 0.024$$

Difference and SE_diff:

$$\text{Difference} = 0.487 - 1.397 = -0.910$$

$$SE_{\text{diff}} = \sqrt{0.781^2 + 0.024^2} \approx \sqrt{0.610 + 0.001} \approx 0.781$$

Z-Score:

$$Z = \frac{-0.910}{0.781} \approx -1.165$$

p-Values:

- $H_1 <$ (Estimate 1 < Estimate 2): $p \approx 0.122$
- $H_1 >$ (Estimate 1 > Estimate 2): $p \approx 0.878$
- $H_1 =$ (Estimate 1 $\neq$ Estimate 2): $p \approx 0.244$

Pair 16

- Estimate 1: 1.103, CI: (0.079, 2.128)
- Estimate 2: 1.466, CI: (1.418, 1.515)

Standard Errors:

$$SE_1 = \frac{2.128 - 0.079}{2 \times 1.96} \approx \frac{2.049}{3.92} \approx 0.523$$

$$SE_2 = \frac{1.515 - 1.418}{2 \times 1.96} \approx \frac{0.097}{3.92} \approx 0.025$$

Difference and SE_diff:

$$\text{Difference} = 1.103 - 1.466 = -0.363$$

$$SE_{\text{diff}} = \sqrt{0.523^2 + 0.025^2} \approx \sqrt{0.274 + 0.001} \approx 0.523$$

Z-Score:

$$Z = \frac{-0.363}{0.523} \approx -0.693$$

p-Values:

- $H_1 <$ (Estimate 1 < Estimate 2): $p \approx 0.245$
- $H_1 >$ (Estimate 1 > Estimate 2): $p \approx 0.755$

- $H_1 = (\text{Estimate 1} \neq \text{Estimate 2}): p \approx 0.490$

Pair 17

- **Estimate 1:** 0.982, CI: (0.063, 1.902)
- **Estimate 2:** 1.472, CI: (1.423, 1.520)

Standard Errors:

$$SE_1 = \frac{1.902 - 0.063}{2 \times 1.96} \approx \frac{1.839}{3.92} \approx 0.468$$

$$SE_2 = \frac{1.520 - 1.423}{2 \times 1.96} \approx \frac{0.097}{3.92} \approx 0.025$$

Difference and SE_diff:

$$\text{Difference} = 0.982 - 1.472 = -0.490$$

$$SE_{\text{diff}} = \sqrt{0.468^2 + 0.025^2} \approx \sqrt{0.219 + 0.001} \approx 0.468$$

Z-Score:

$$Z = \frac{-0.490}{0.468} \approx -1.047$$

p-Values:

- $H_1 < (\text{Estimate 1} < \text{Estimate 2}): p \approx 0.146$
- $H_1 > (\text{Estimate 1} > \text{Estimate 2}): p \approx 0.854$
- $H_1 = (\text{Estimate 1} \neq \text{Estimate 2}): p \approx 0.292$

Pair 18

- **Estimate 1:** 0.893, CI: (0.044, 1.742)
- **Estimate 2:** 1.528, CI: (1.484, 1.571)

Standard Errors:

$$SE_1 = \frac{1.742 - 0.044}{2 \times 1.96} \approx \frac{1.698}{3.92} \approx 0.433$$

$$SE_2 = \frac{1.571 - 1.484}{2 \times 1.96} \approx \frac{0.087}{3.92} \approx 0.022$$

Difference and SE_diff:

$$\text{Difference} = 0.893 - 1.528 = -0.635$$

$$SE_{\text{diff}} = \sqrt{0.433^2 + 0.022^2} \approx \sqrt{0.187 + 0.000} \approx 0.433$$

Z-Score:

$$Z = \frac{-0.635}{0.433} \approx -1.465$$

p-Values:

- $H_1 <$ (Estimate 1 < Estimate 2): $p \approx 0.071$
- $H_1 >$ (Estimate 1 > Estimate 2): $p \approx 0.929$
- $H_1 =$ (Estimate 1 $\neq$ Estimate 2): $p \approx 0.143$

Pair 19

- Estimate 1: 0.730, CI: (0.038, 1.422)
- Estimate 2: 1.525, CI: (1.474, 1.575)

Standard Errors:

$$SE_1 = \frac{1.422 - 0.038}{2 \times 1.96} \approx \frac{1.384}{3.92} \approx 0.353$$

$$SE_2 = \frac{1.575 - 1.474}{2 \times 1.96} \approx \frac{0.101}{3.92} \approx 0.026$$

Difference and SE_diff:

$$\text{Difference} = 0.730 - 1.525 = -0.795$$

$$SE_{\text{diff}} = \sqrt{0.353^2 + 0.026^2} \approx \sqrt{0.125 + 0.001} \approx 0.353$$

Z-Score:

$$SE_1 = \frac{1.422 - 0.038}{2 \times 1.96} \approx \frac{1.384}{3.92} \approx 0.353$$

$$SE_2 = \frac{1.575 - 1.474}{2 \times 1.96} \approx \frac{0.101}{3.92} \approx 0.026$$

Difference and SE_diff:

$$\text{Difference} = 0.730 - 1.525 = -0.795$$

$$SE_{\text{diff}} = \sqrt{0.353^2 + 0.026^2} \approx \sqrt{0.125 + 0.001} \approx 0.353$$

Z-Score:

$$Z = \frac{-0.795}{0.353} \approx -2.252$$

p-Values:

- $H_1 < (\text{Estimate 1} < \text{Estimate 2}): p \approx 0.012$
- $H_1 > (\text{Estimate 1} > \text{Estimate 2}): p \approx 0.988$
- $H_1 = (\text{Estimate 1} \neq \text{Estimate 2}): p \approx 0.024$

Pair 20

- **Estimate 1:** 0.581, CI: (0.234, 0.928)
- **Estimate 2:** 1.482, CI: (1.387, 1.587)

Standard Errors:

$$SE_1 = \frac{0.928 - 0.234}{2 \times 1.96} \approx \frac{0.694}{3.92} \approx 0.177$$

$$SE_2 = \frac{1.587 - 1.387}{2 \times 1.96} \approx \frac{0.200}{3.92} \approx 0.051$$

Difference and SE_diff:

$$\text{Difference} = 0.581 - 1.482 = -0.901$$

$$SE_{\text{diff}} = \sqrt{0.177^2 + 0.051^2} \approx \sqrt{0.031 + 0.003} \approx 0.179$$

Z-Score:

$$Z = \frac{-0.901}{0.179} \approx -5.036$$

p-Values:

- $H_1 <$ (Estimate 1 < Estimate 2): $p < 0.001$
- $H_1 >$ (Estimate 1 > Estimate 2): $p \approx 1.000$
- $H_1 =$ (Estimate 1 $\neq$ Estimate 2): $p < 0.001$